Banking Customer Engagement & Deposit Propensity Analytics

Which prospects deserve the next term-deposit conversation?

Limited outreach capacity makes prioritization useful.
The system ranks historical records by subscription score.

Campus deck connection

Customer engagement, Customer 360, and sales-oriented relationships

Human decision

The relationship manager checks customer needs and product suitability.

Data and project pipeline

41,188 historical campaign records

UCI bank-additional-full.csv, Portugal, May 2008 to November 2010

Stage	What the project does
1 Ingest and audit	Preserve order, unknown values, and surrogate record IDs
2 SQLite and SQL	Describe customer segments and prior campaign outcomes
3 Model training	Training-only preprocessing and two model families
4 Score and review	Score records, apply simulated rules, and export a shortlist
5 Streamlit demo	Explore history, model results, outreach replay, and scoring

Telephone campaign history. Digital-channel examples and engagement flags are simulated.

Leakage controls and ordered evaluation

Partition	Rows	Purpose
Earliest 70%	28,831	Fit features and model candidates
Next 15%	6,178	Choose family finalists, champion, and cutoff
Final 15%	6,179	Evaluate frozen choices once

Scoring happens before the proposed contact

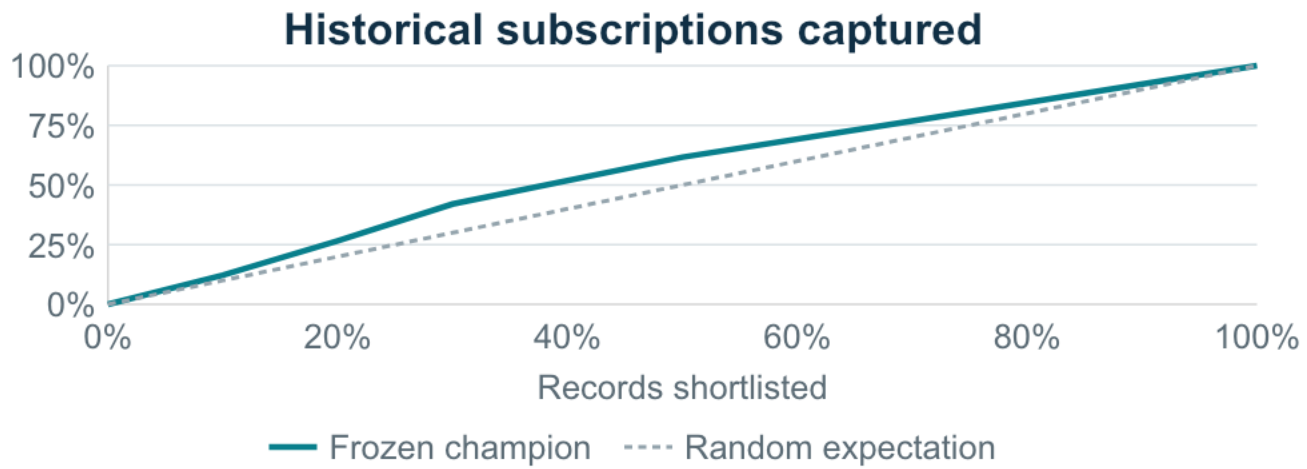
Exclude duration and uncertain current-contact fields.
Keep macro indicators out of the conservative model.

Freeze on validation Precision@20%, then AP.
Prevalence: 5.6% train, 10.7% validation, 38.5% test.

Measured results on the final chronological holdout

Test: 6,179 records, 2,376 subscriptions, 38.5% base rate

Model	PR-AUC	AP	ROC-AUC	Precision @20%	Lift @20%	Recall @20%
Prior baseline	0.692	0.385	0.500	37.6%	0.98×	19.6%
Logistic Regression	0.522	0.523	0.622	60.0%	1.56×	31.2%
Random Forest	**0.474**	**0.474**	**0.619**	**51.1%**	**1.33×**	**26.6%**



Random Forest
Validation-selected champion

632 of 2,376 subscriptions in the top 1,236 records

Offline historical capture.
No causal sales or revenue claim.

Relationship-manager demo and judgment

Demo action	What the relationship manager sees
Set outreach capacity	A ranked historical replay pool with clear record counts
Apply eligibility rules	Simulated opt-outs suppress outreach regardless of score
Review an eligible record	Model score plus recorded prior-contact context
Download the shortlist	Surrogate IDs and context, without outcomes or contact details

Conversation prompt

“Would a fixed-term savings option fit your liquidity needs and financial goals?”

Future use needs current data, real permissions, and a controlled evaluation of incremental benefit.